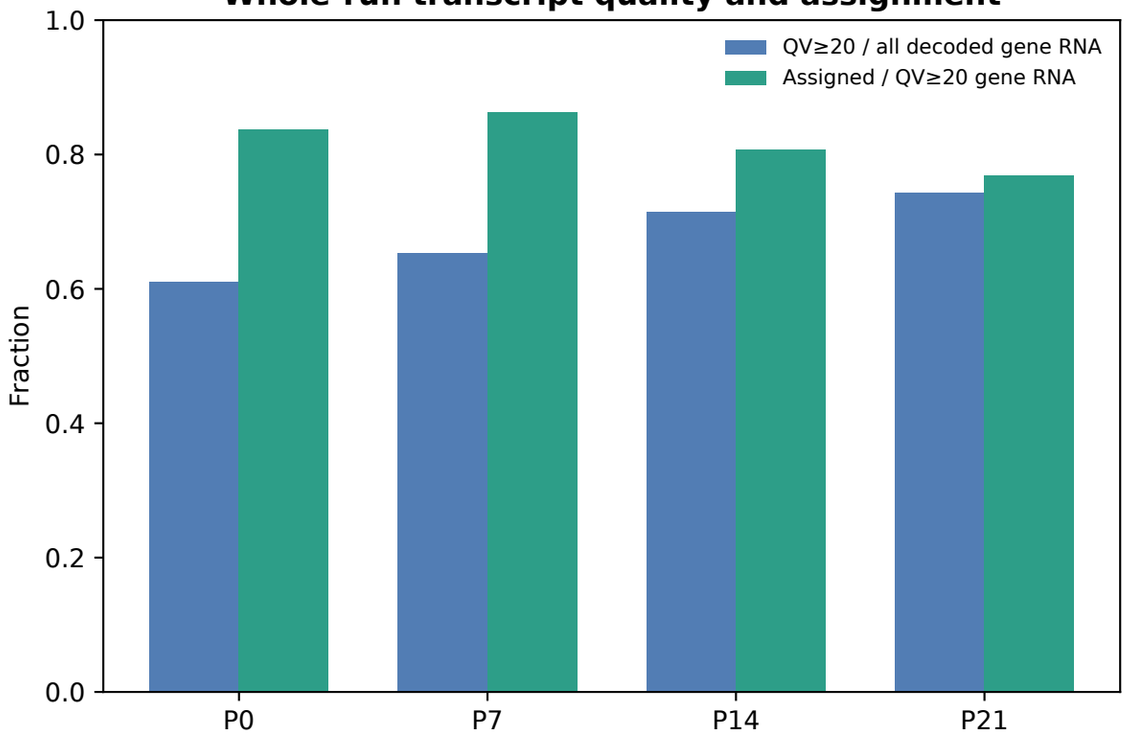
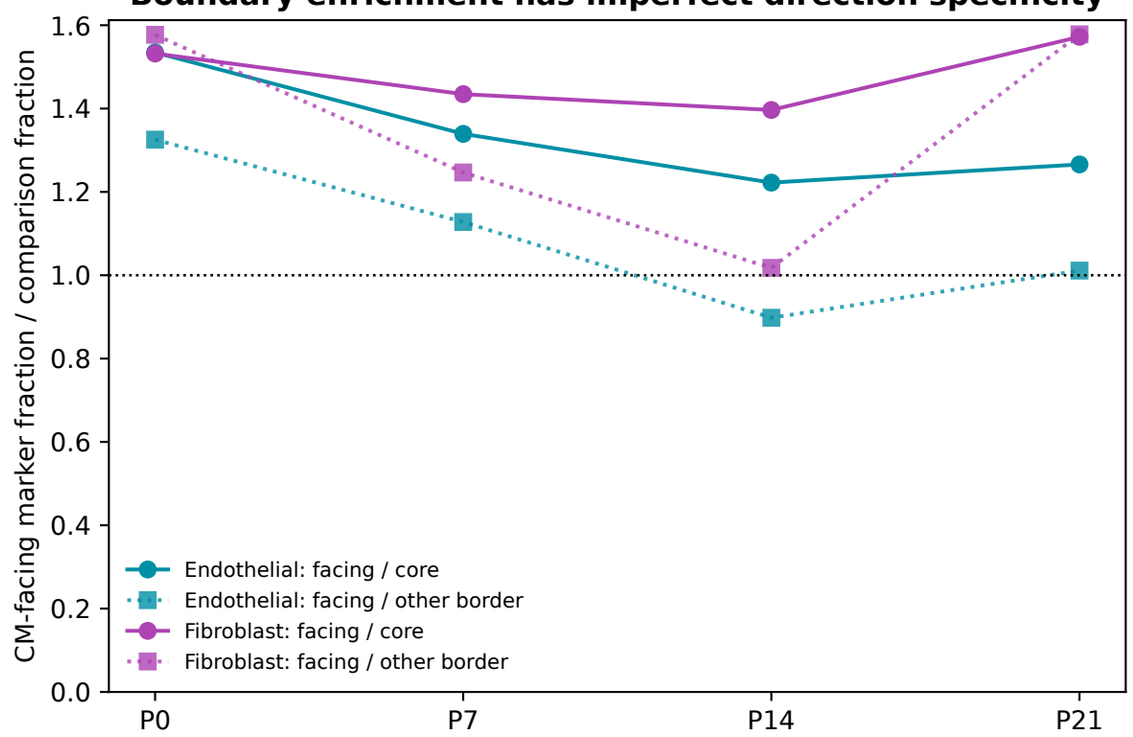
Xenium transcript-boundary audit and segmentation sensitivity

Whole-run transcript quality and assignment



Boundary enrichment has imperfect direction specificity



All 3,104 original matched CM pairs

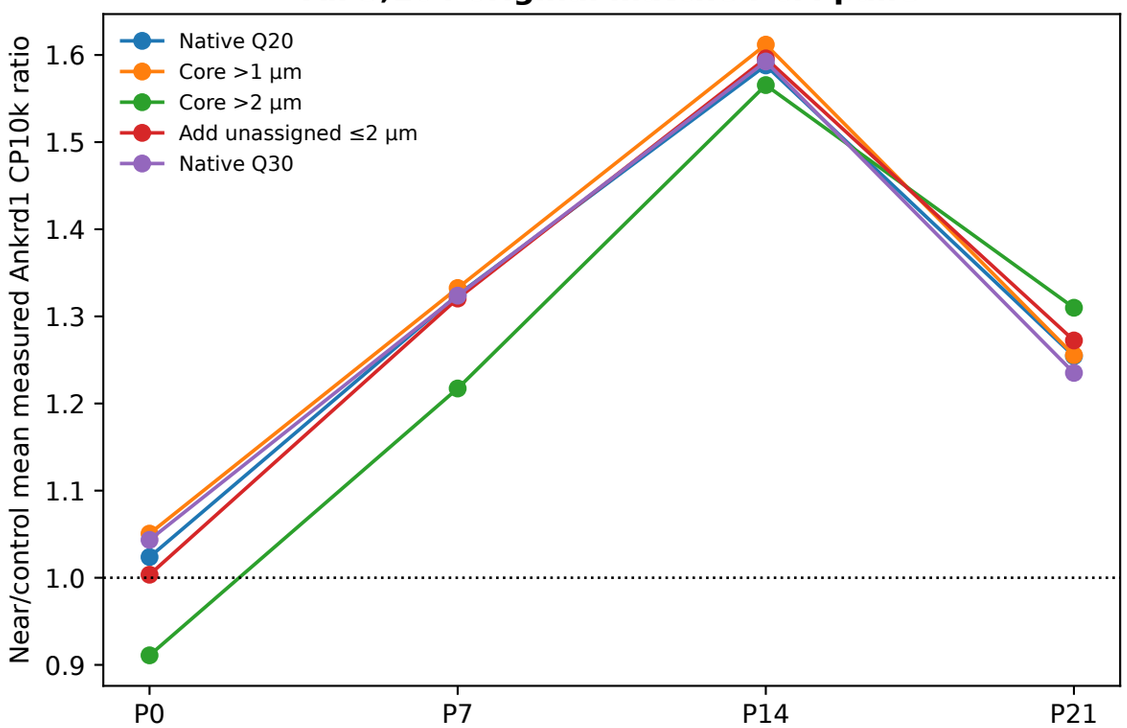
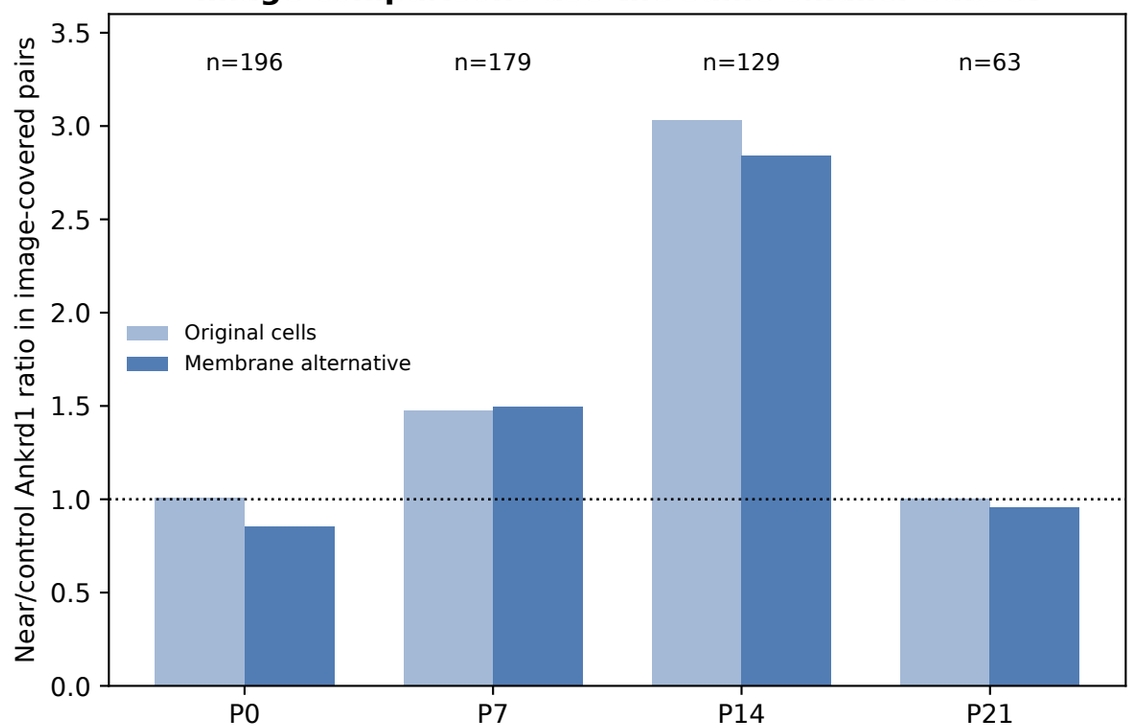


Image comparisons use the same smaller subset



Molecule reconstruction matches every deposited count matrix exactly at QV≥20. No nucleus is required for a CM profile.
Boundary-localized marker RNA is compatible with mixing, not proof of contamination. Alternative segmentation is not validated ground truth.
One section per stage; ratios are within-section contrasts. Full-pair and image-region subsets must not be treated as the same cohort.

Broad-identity stability depends on boundaries and measurement depth

Boundary/QV changes and count-matched controls

	Core >1 µm	Random: core1 depth	Core >2 µm	Random: core2 depth	Add unassigned ≤2 µm	Native Q30	Random: Q30 depth
P0 · CM (n=5864)	86%	86%	54%	56%	98%	92%	95%
P0 · EC (n=1312)	77%	73%	49%	40%	93%	78%	79%
P0 · Fib (n=491)	68%	62%	35%	28%	95%	64%	68%
P0 · SMC (n=35)	57%	46%	29%	23%	91%	57%	57%
P7 · CM (n=7025)	91%	91%	66%	66%	98%	94%	96%
P7 · EC (n=1541)	68%	57%	33%	19%	94%	71%	71%
P7 · Fib (n=470)	64%	59%	32%	24%	94%	56%	65%
P7 · SMC (n=44)	50%	48%	20%	9%	98%	52%	52%
P14 · CM (n=5032)	95%	95%	83%	83%	97%	98%	98%
P14 · EC (n=1284)	70%	62%	40%	27%	90%	81%	81%
P14 · Fib (n=437)	69%	65%	40%	36%	88%	78%	81%
P14 · SMC (n=122)	73%	70%	43%	38%	90%	71%	82%
P21 · CM (n=3212)	91%	92%	76%	76%	97%	97%	97%
P21 · EC (n=1136)	65%	56%	32%	21%	85%	79%	81%
P21 · Fib (n=661)	69%	69%	42%	36%	92%	84%	84%
P21 · SMC (n=71)	63%	63%	30%	25%	85%	79%	83%

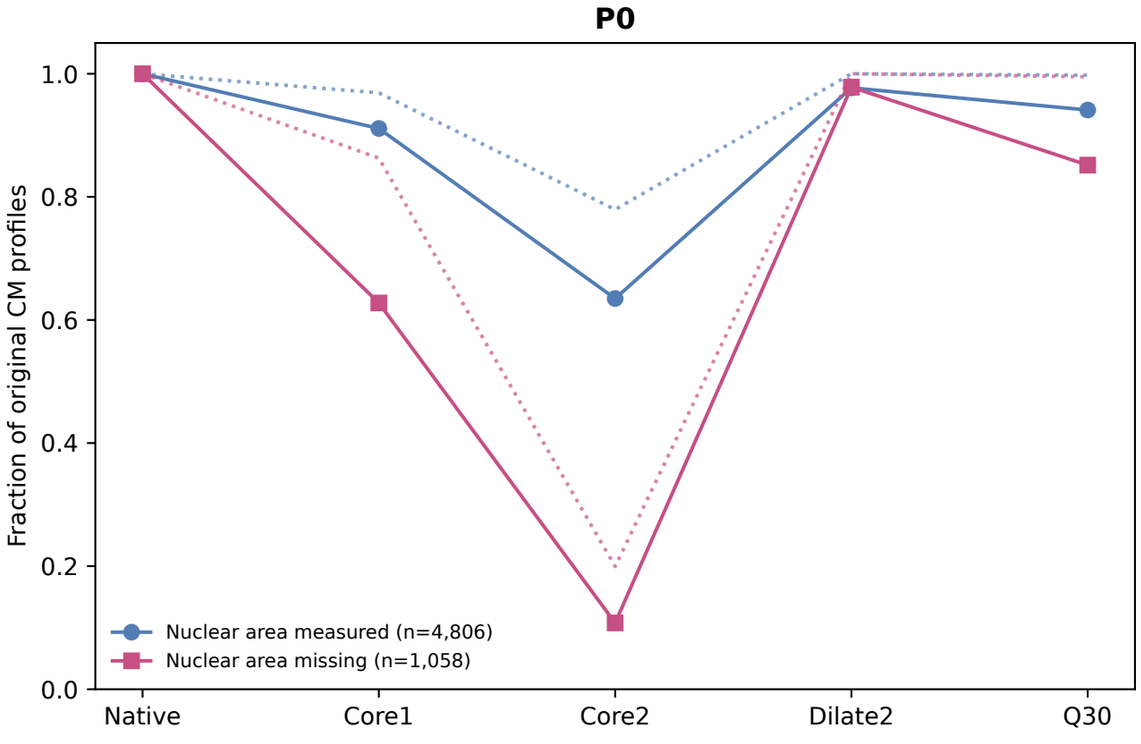
Image-covered cells: boundary alternatives

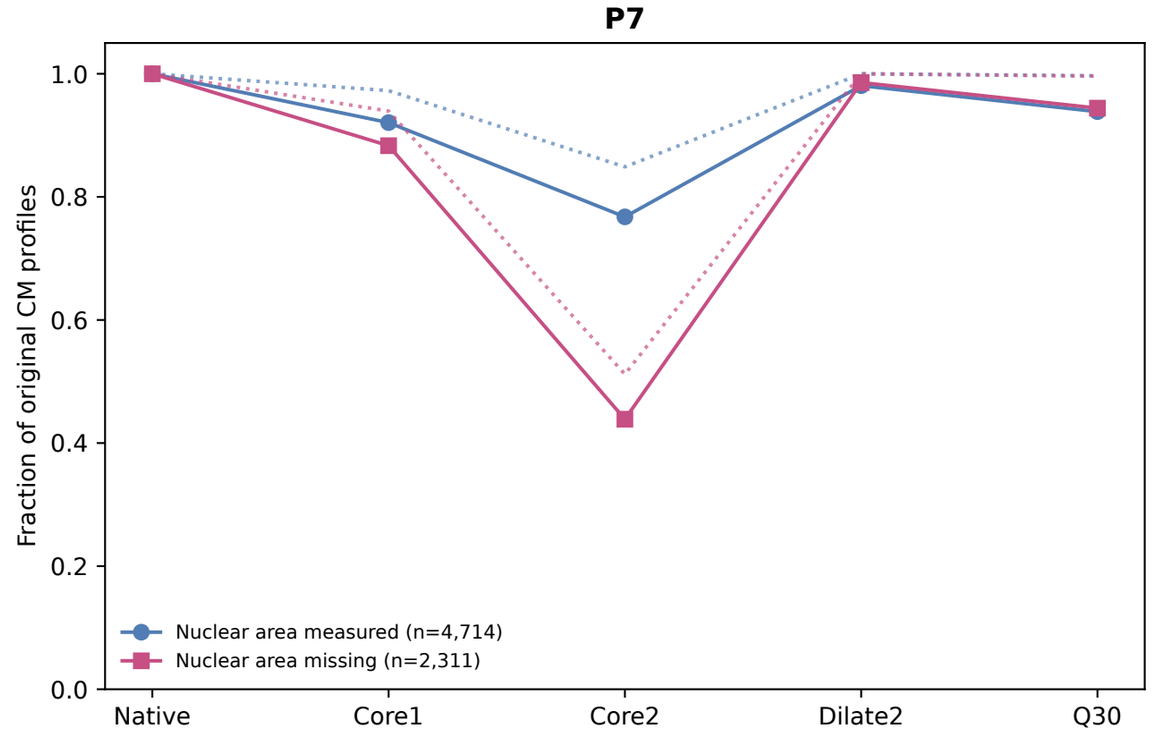
	Native raster control	Centroid Voronoi	Membrane watershed
P0 · CM (n=3422)	99%	93%	86%
P0 · EC (n=833)	95%	87%	71%
P0 · Fib (n=341)	93%	87%	64%
P0 · SMC (n=25)	88%	84%	48%
P7 · CM (n=3393)	99%	95%	89%
P7 · EC (n=1009)	95%	83%	67%
P7 · Fib (n=294)	94%	78%	54%
P7 · SMC (n=38)	92%	76%	55%
P14 · CM (n=2076)	99%	95%	88%
P14 · EC (n=854)	94%	78%	60%
P14 · Fib (n=312)	94%	76%	55%
P14 · SMC (n=95)	93%	80%	63%
P21 · CM (n=1613)	98%	95%	86%
P21 · EC (n=695)	94%	64%	52%
P21 · Fib (n=415)	96%	80%	64%
P21 · SMC (n=46)	93%	72%	59%

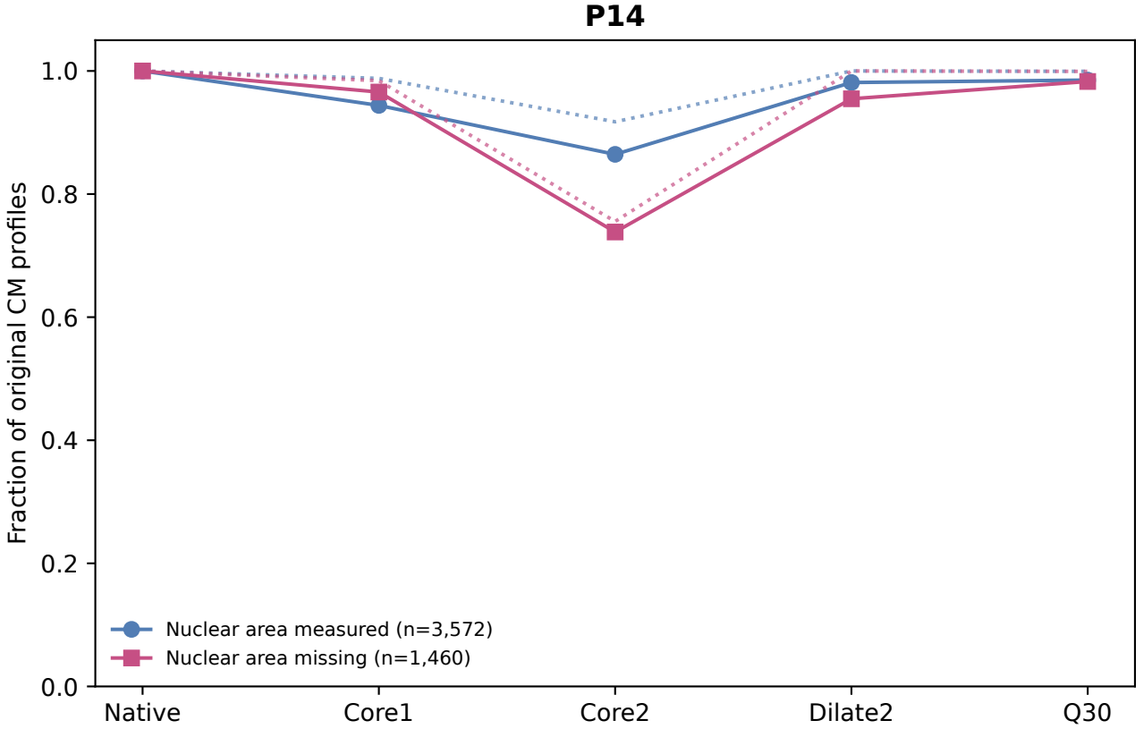
Fraction retaining the original broad label

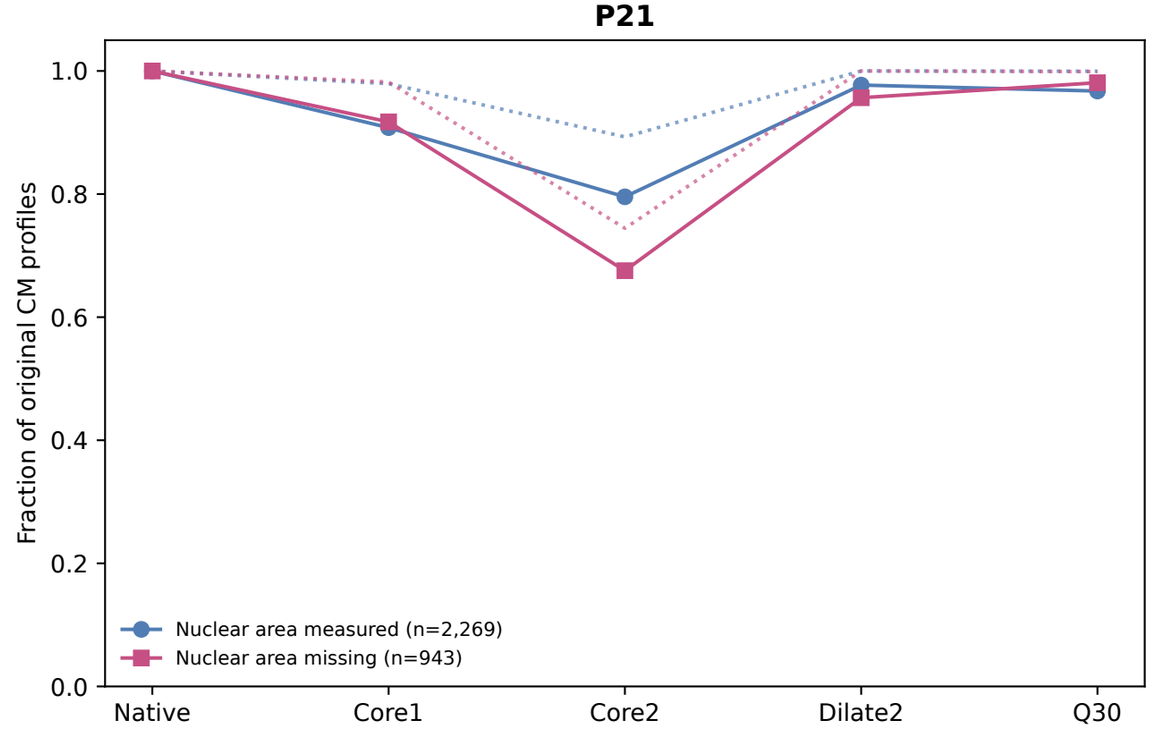
Label retention is not an accuracy estimate or segmentation-error rate. Lost labels include uncertainty and insufficient RNA, not only lineage switches.
Random controls sample original transcripts without replacement to exactly match each cell's eroded/Q30 transcript total (one fixed technical draw).
The two panels use different cohorts; image comparisons require adequate crop coverage. All original seeds, including anucleate profiles, are retained.

CM profiles without measured nuclear area are retained

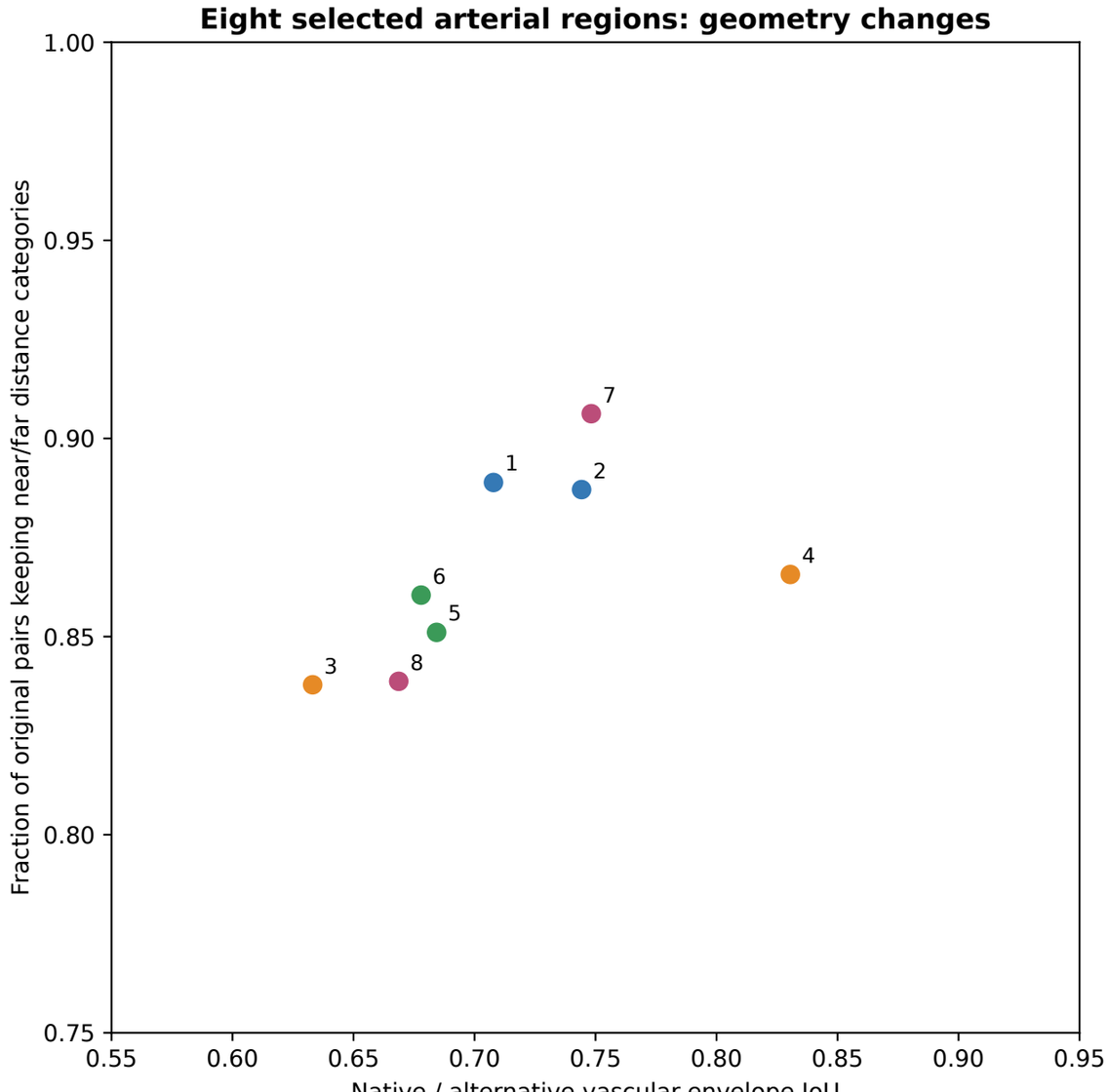


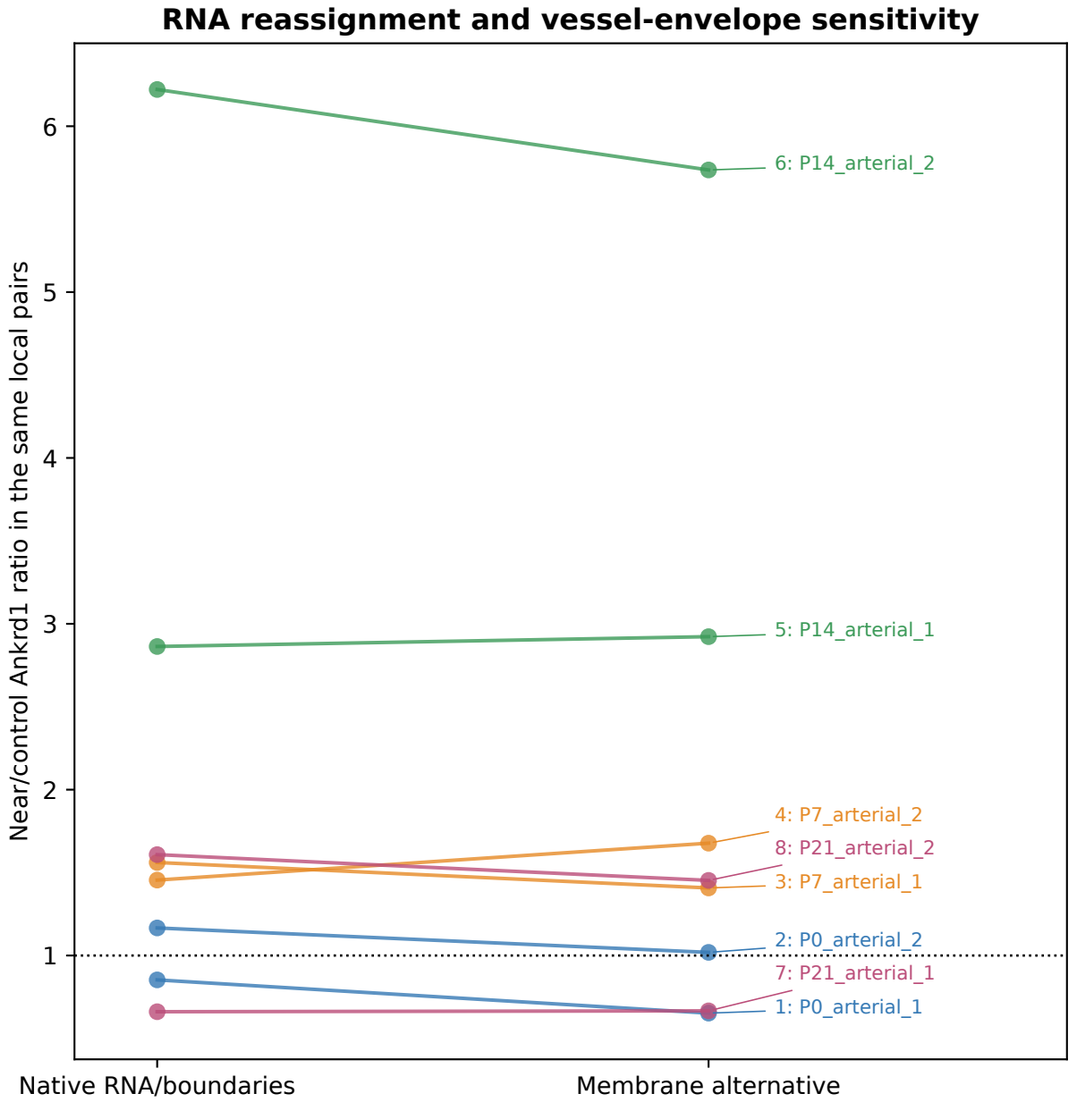






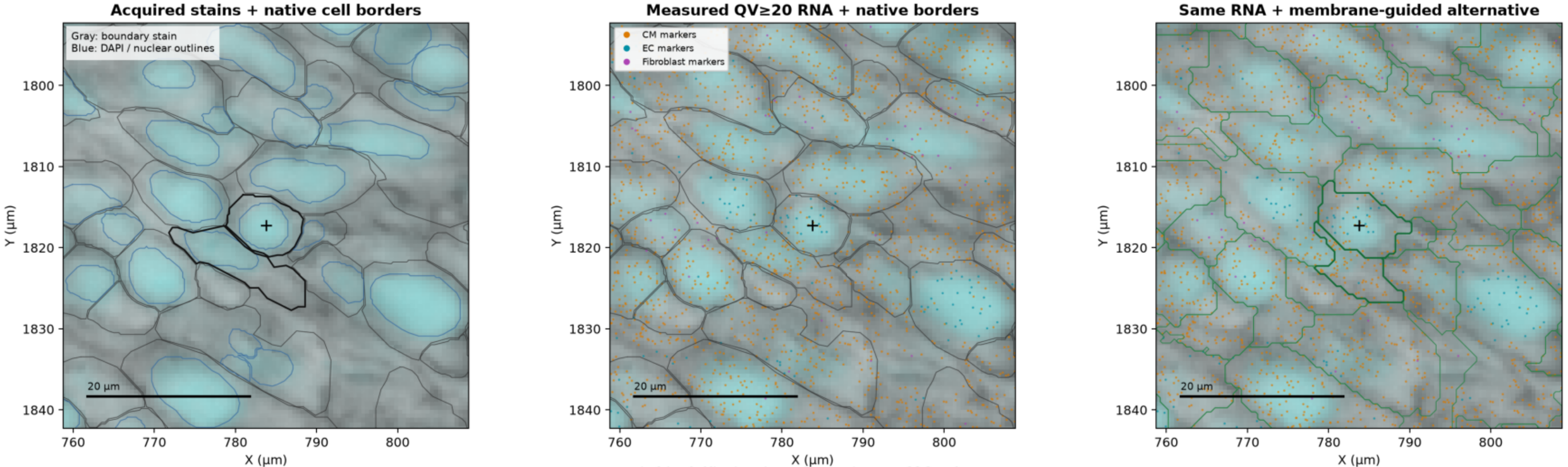
Solid: same CM identity; dotted: sufficient expression information. No nucleus criterion enters classification or alternative seed placement.
Large RNA losses under erosion are stress tests, not recommendations to discard legitimate anucleate CM profiles.
Nuclear-area groups differ biologically and technically; these curves do not estimate a causal effect of nuclear presence.





Same original arterial EC/SMC memberships; envelopes rebuilt from each alternative's cell polygons, buffered3 µm.
Comparisons use crop-covered original pairs. Additional distance-preserved-pair results are in the table.
These are eight selected vessel regions, not eight independent animals or a complete reannotation of the vasculature.

P0_Endothelial_flagged: CM–Endothelial interface

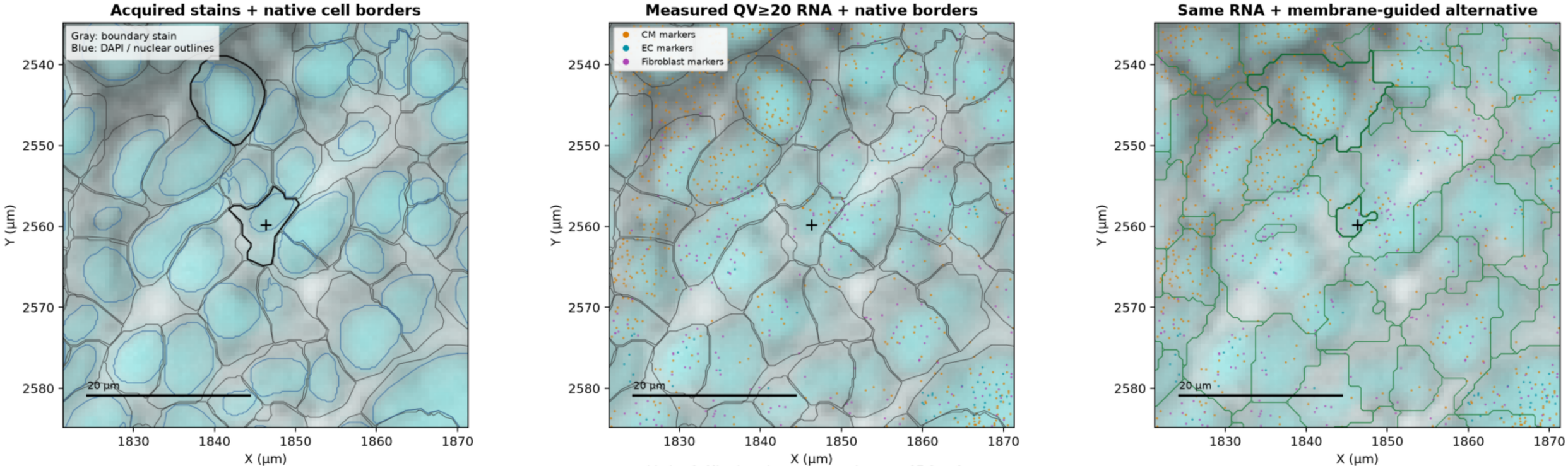


abnfoiao-1: Mixed-marker review; nuclear area 28.2 µm²

Every displayed marker molecule retains its original position; no expression smoothing.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P0_Fibroblast_flagged: CM–Fibroblast interface

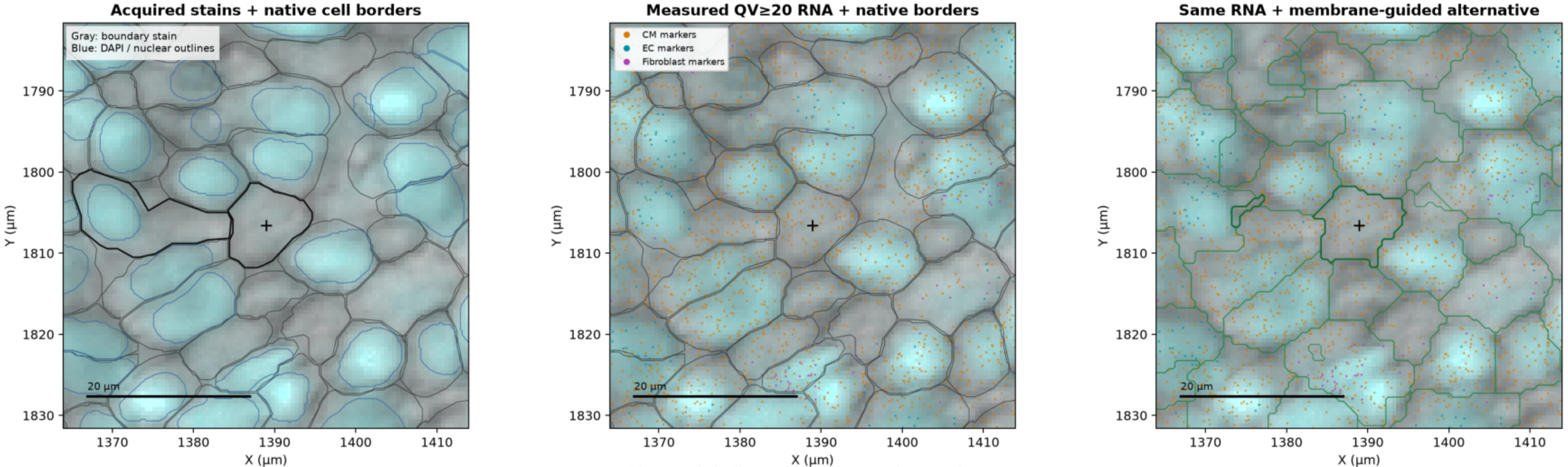


mebhojgo-1: Mixed-marker review; nuclear area 17.6 µm²

Every displayed marker molecule retains its original position; no expression smoothing.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P0_anucleate_CM_1: CM without measured nuclear area

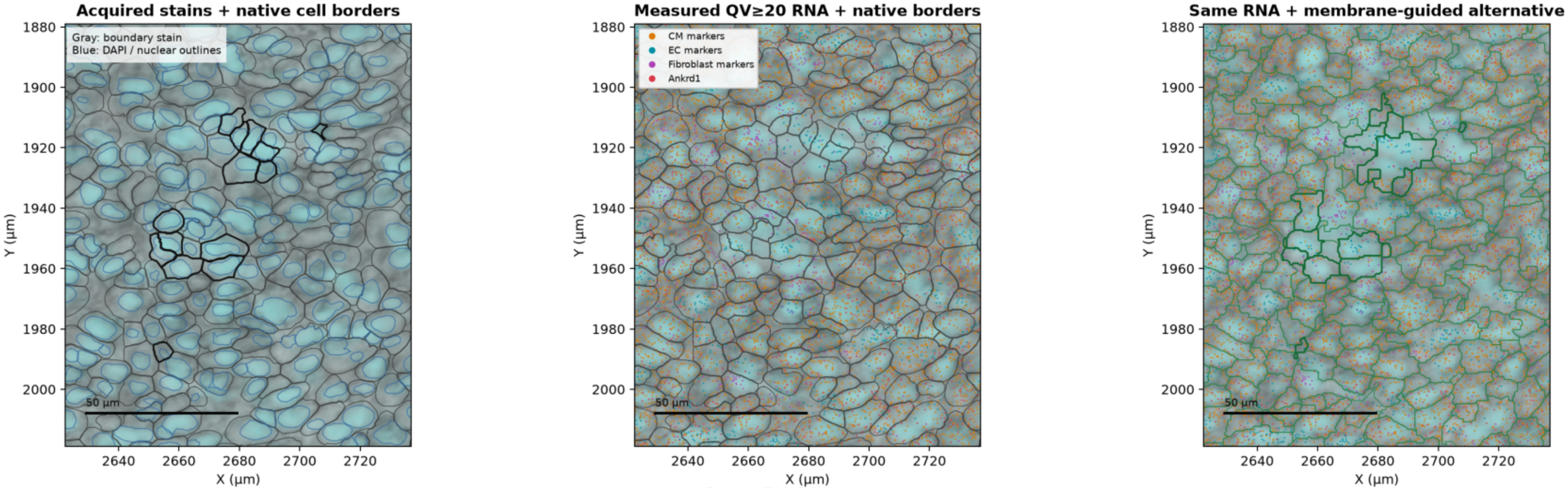


bbcnoama-1: Cardiomyocyte; nuclear area not measured

Every displayed marker molecule retains its original position; no expression smoothing.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P0_arterial_1: Peri-arterial

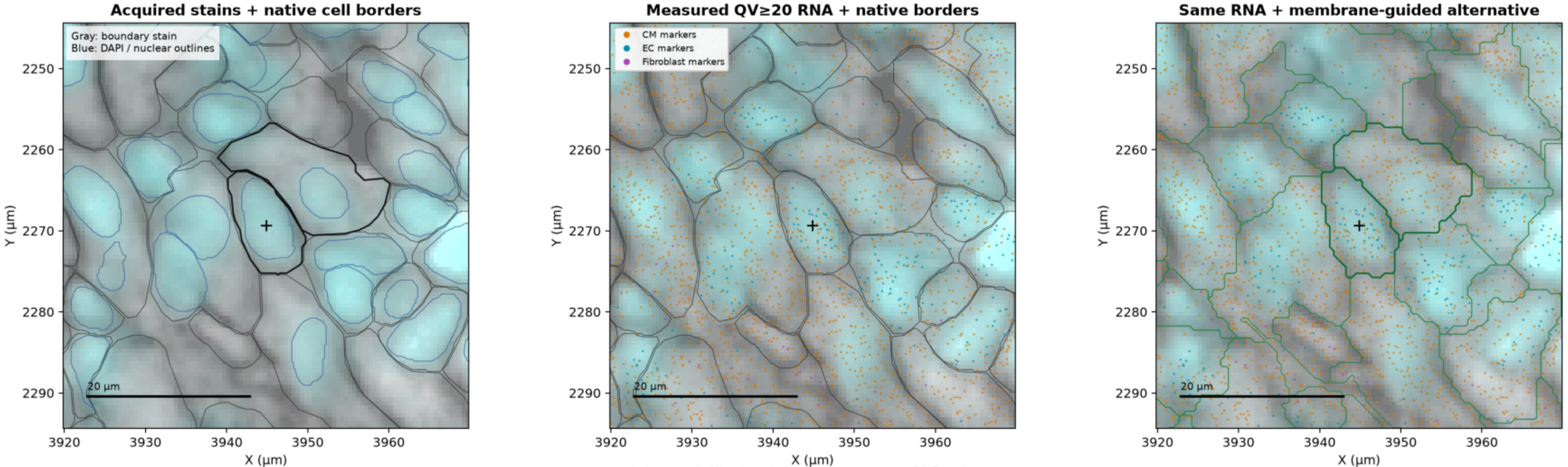


7 arterial EC + 7 SMC seeds

RNA points sampled to≤5,000 per marker group for display; all molecules retained in data.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P7_Endothelial_flagged: CM–Endothelial interface

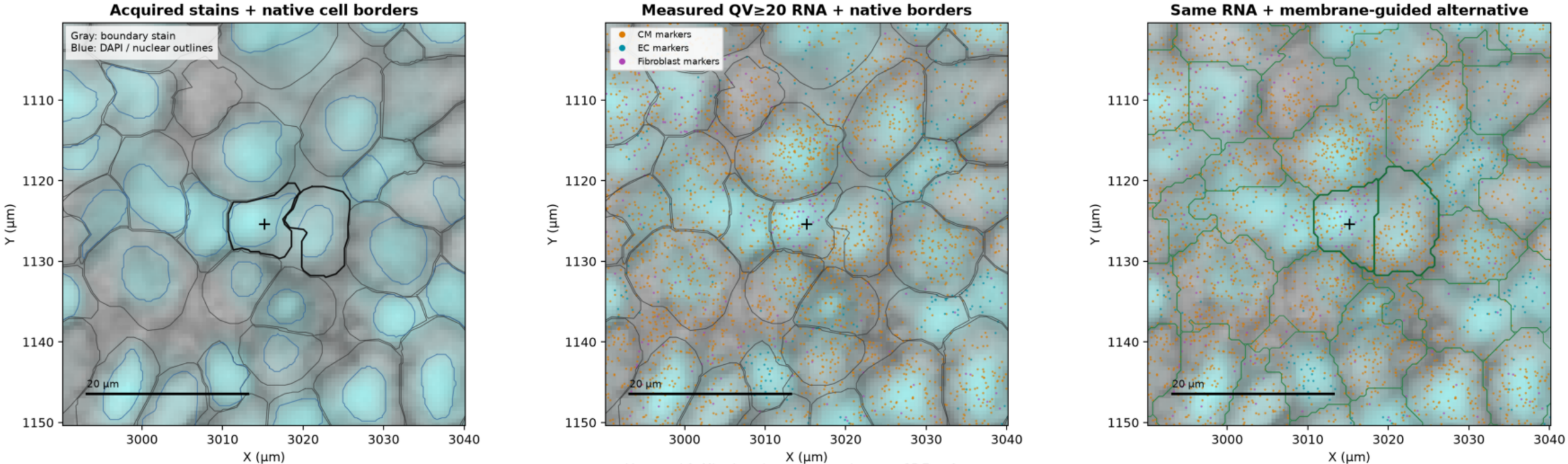


dmkempck-1: Mixed-marker review; nuclear area 50.7 µm²

Every displayed marker molecule retains its original position; no expression smoothing.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P7_Fibroblast_flagged: CM–Fibroblast interface

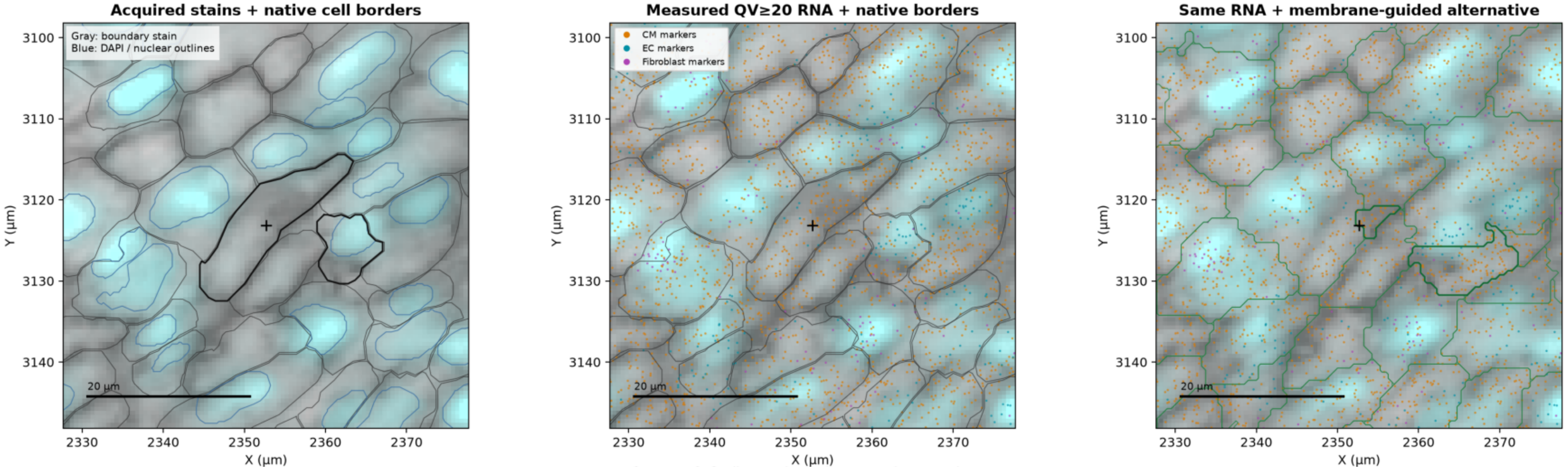


kbeappgd-1: Mixed-marker review; nuclear area 35.7 µm²

Every displayed marker molecule retains its original position; no expression smoothing.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P7_anucleate_CM_1: CM without measured nuclear area

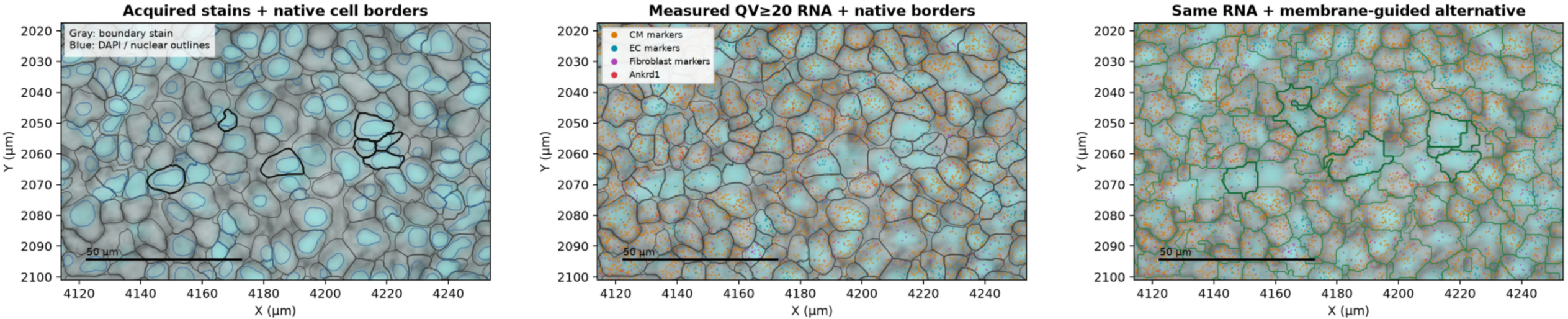


facpamoa-1: Cardiomyocyte; nuclear area not measured

Every displayed marker molecule retains its original position; no expression smoothing.

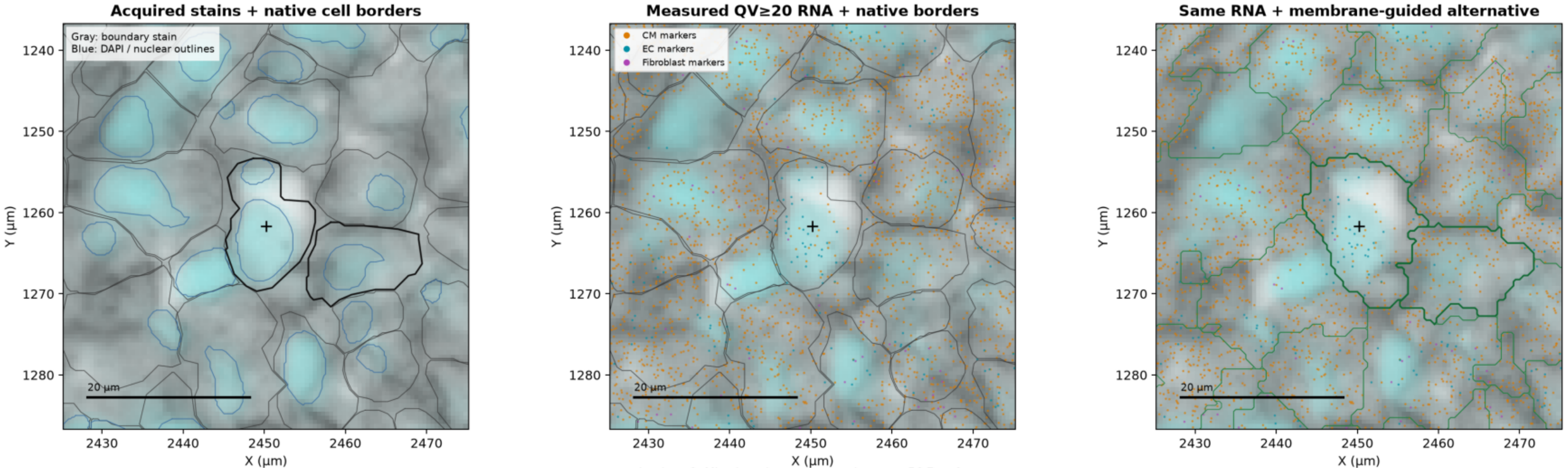
Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P7_arterial_1: Peri-arterial



4 arterial EC + 3 SMC seeds
RNA points sampled to≤5,000 per marker group for display; all molecules retained in data.
Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P14_Endothelial_flagged: CM–Endothelial interface

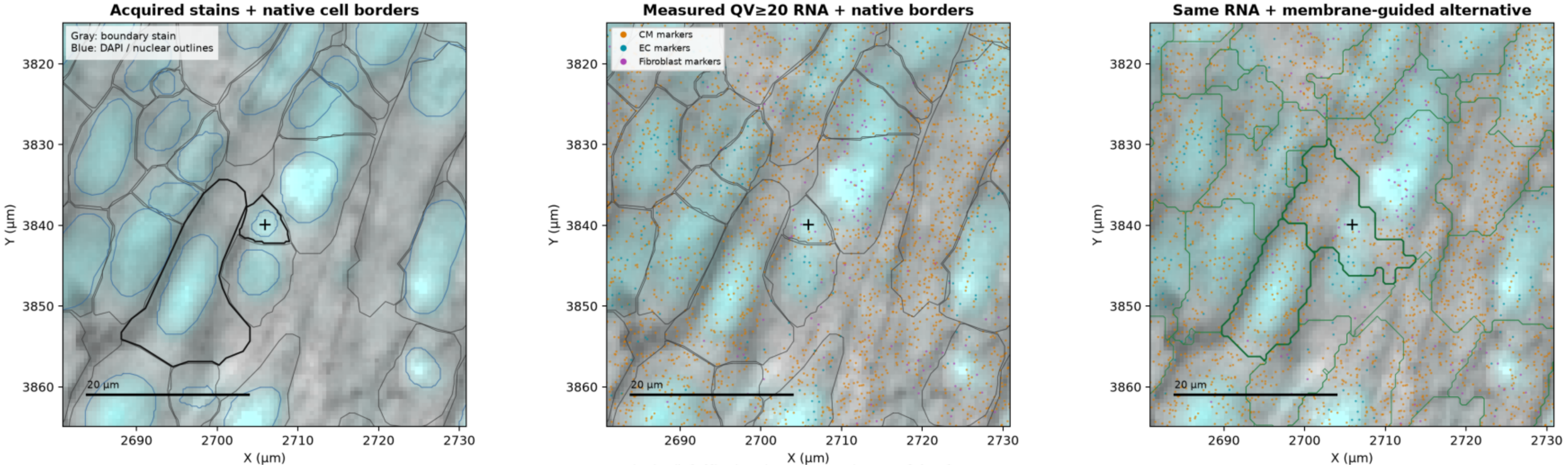


ebpgjecn-1: Mixed-marker review; nuclear area 56.7 µm²

Every displayed marker molecule retains its original position; no expression smoothing.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P14_Fibroblast_flagged: CM–Fibroblast interface

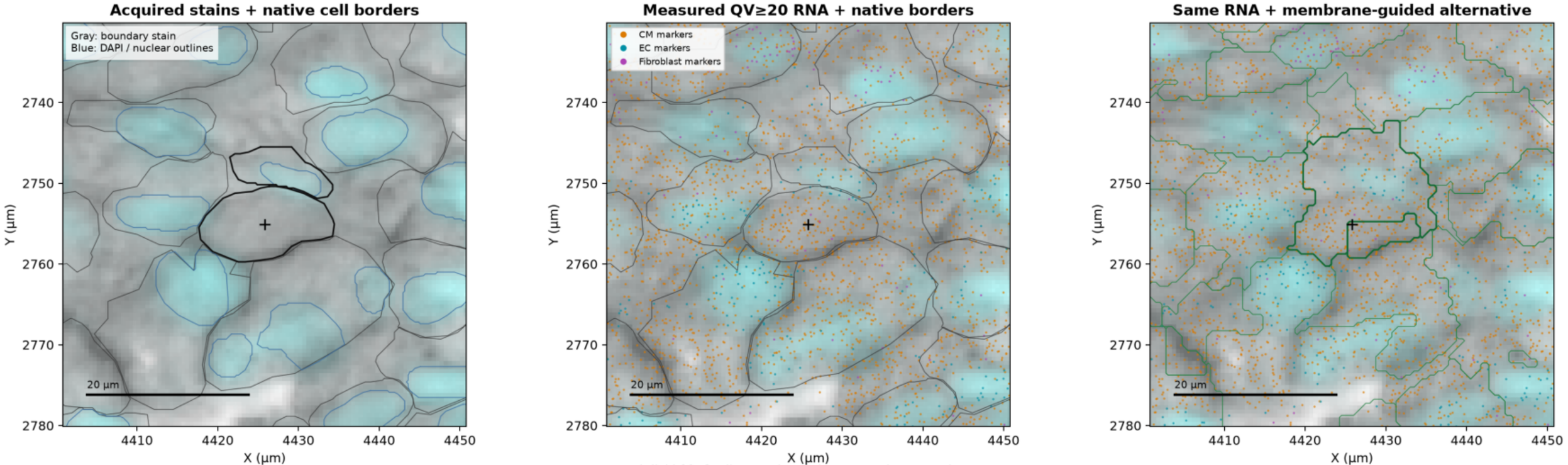


kaeimpih-1: Mixed-marker review; nuclear area 8.4 µm²

Every displayed marker molecule retains its original position; no expression smoothing.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P14_anucleate_CM_1: CM without measured nuclear area

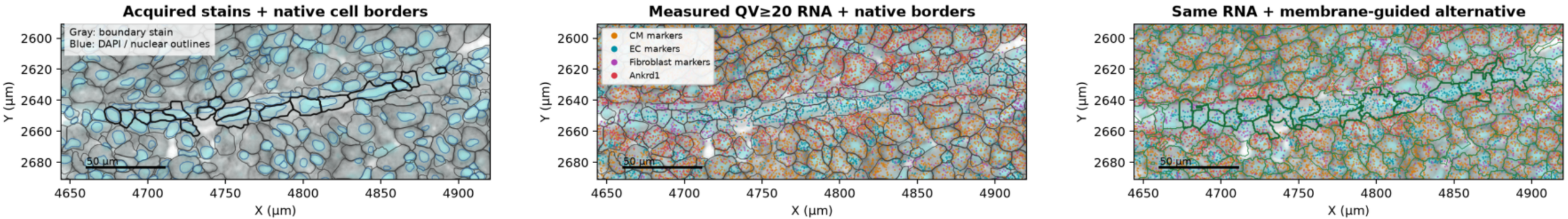


ebdlgbbf-1: Cardiomyocyte; nuclear area not measured

Every displayed marker molecule retains its original position; no expression smoothing.

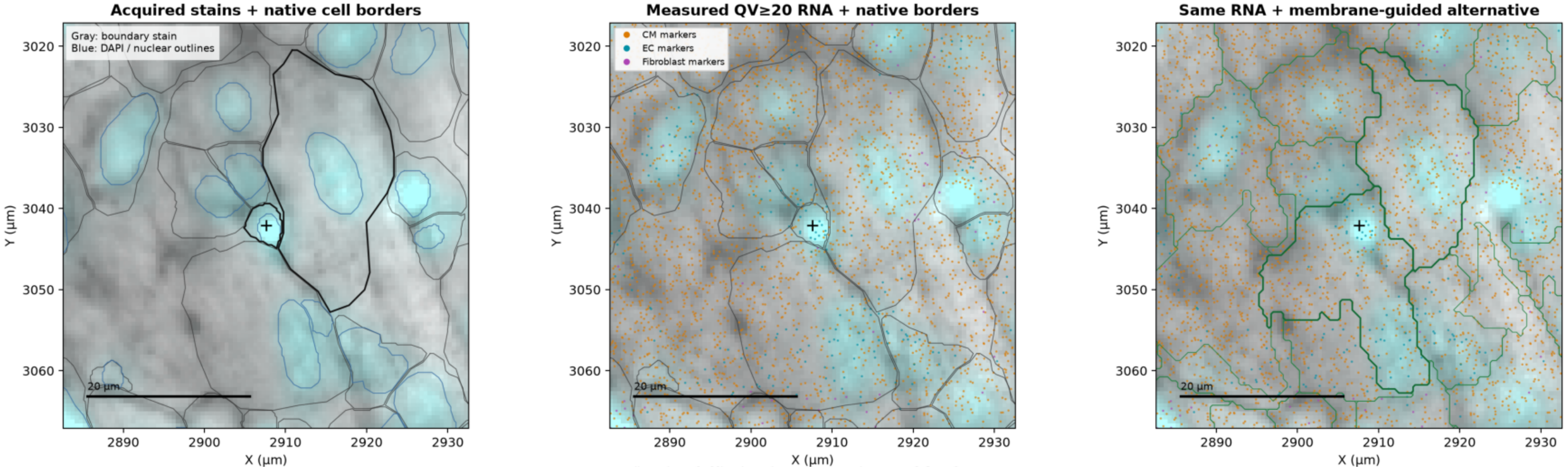
Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P14_arterial_1: Peri-arterial



10 arterial EC + 7 SMC seeds
RNA points sampled to≤5,000 per marker group for display; all molecules retained in data.
Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P21_Endothelial_flagged: CM–Endothelial interface

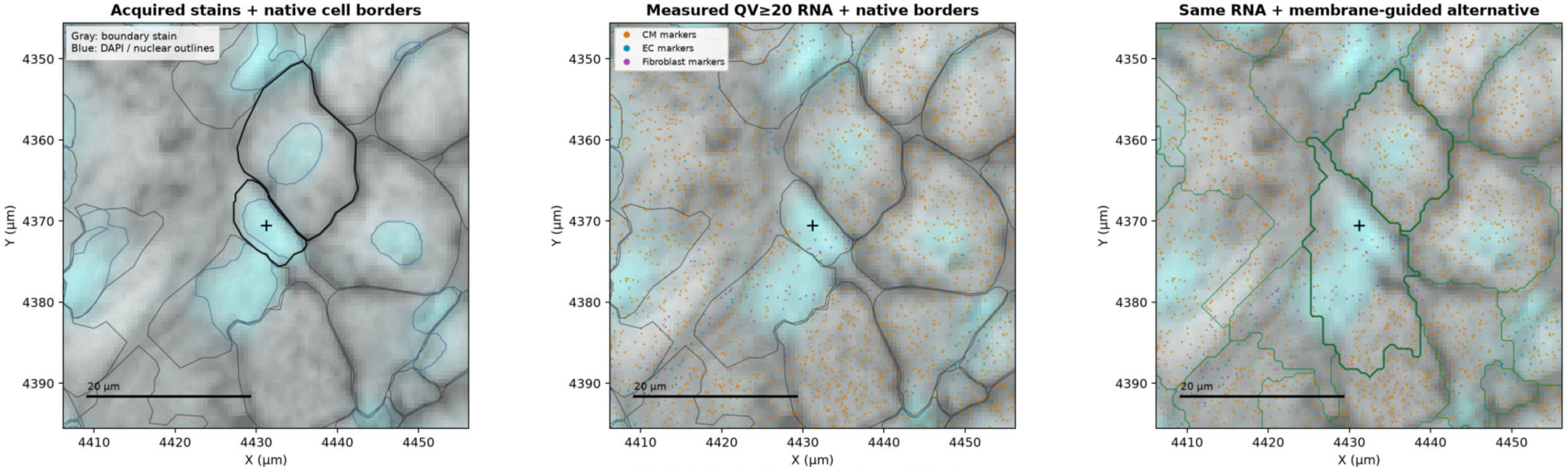


jhmmjcpa-1: Mixed-marker review; nuclear area 9.8 µm²

Every displayed marker molecule retains its original position; no expression smoothing.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P21_Fibroblast_flagged: CM–Fibroblast interface

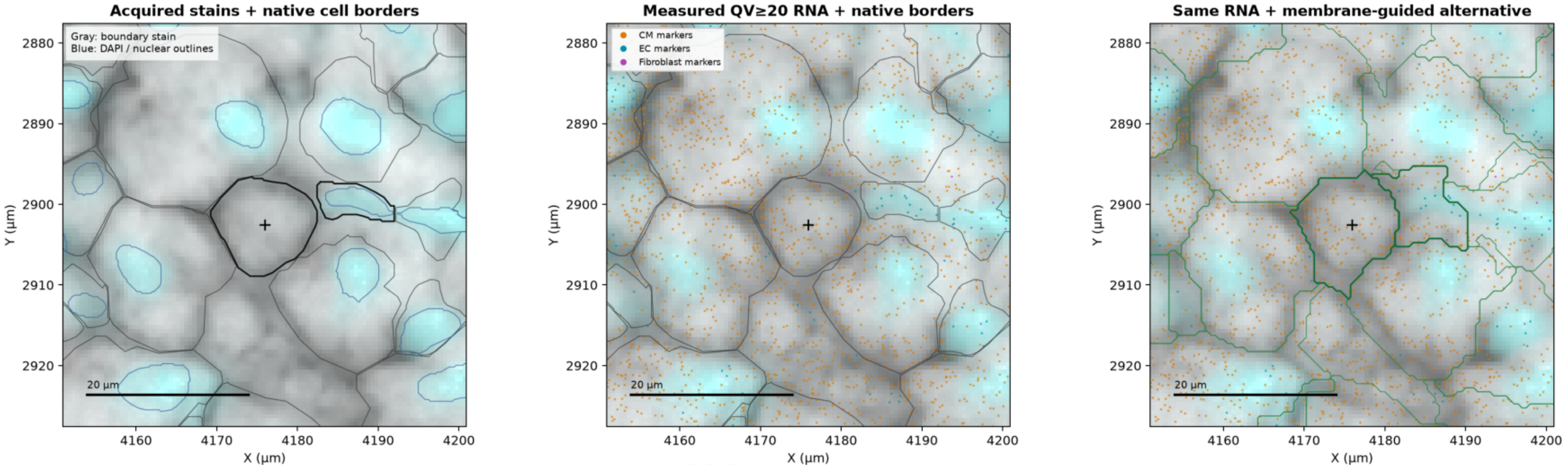


bglabdbe-1: Mixed-marker review; nuclear area 33.1 µm²

Every displayed marker molecule retains its original position; no expression smoothing.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P21_anucleate_CM_1: CM without measured nuclear area

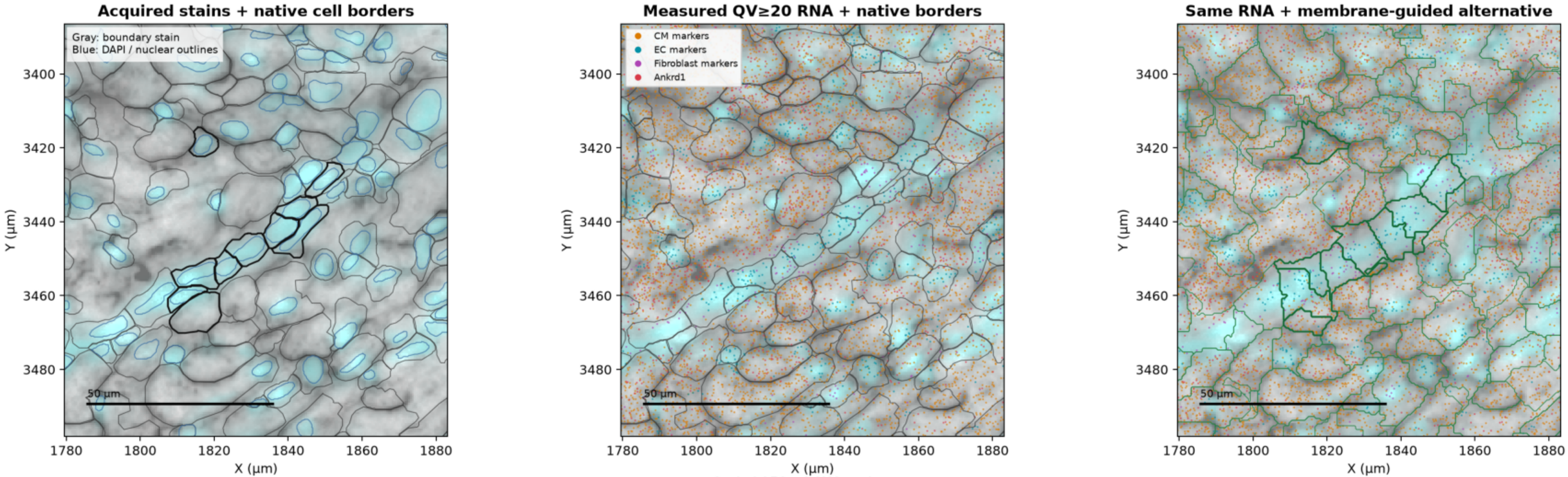


akpncpcc-1: Cardiomyocyte; nuclear area not measured

Every displayed marker molecule retains its original position; no expression smoothing.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.

P21_arterial_1: Peri-arterial



2 arterial EC + 8 SMC seeds

RNA points sampled to≤5,000 per marker group for display; all molecules retained in data.

Alternative: fixed original cell seeds, including anucleate profiles; boundary stain only. This is a sensitivity test, not validated replacement segmentation.